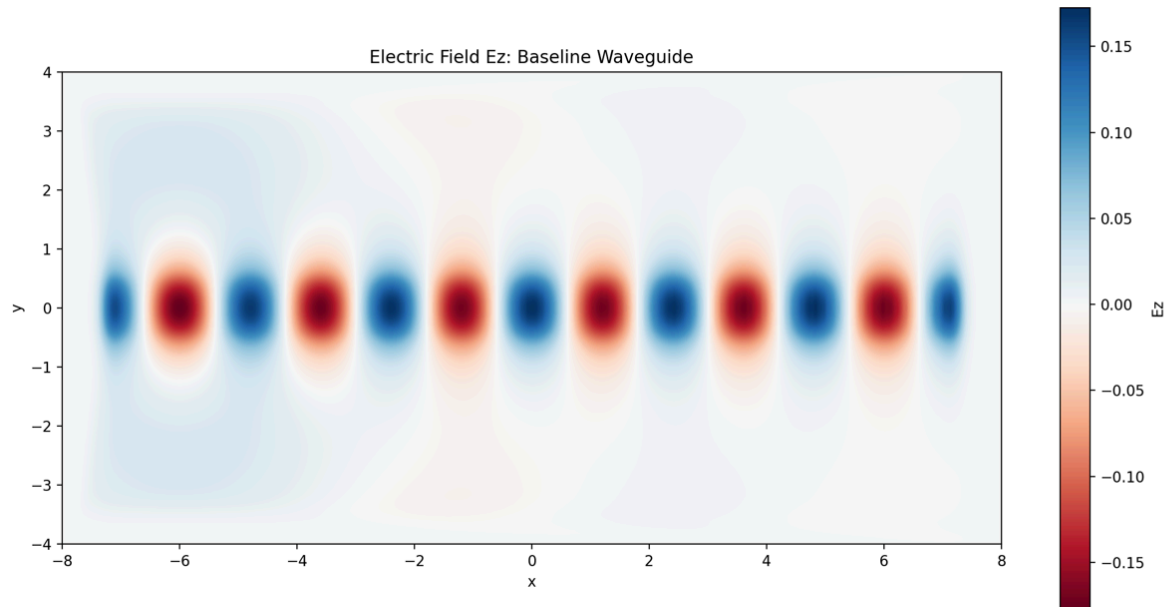


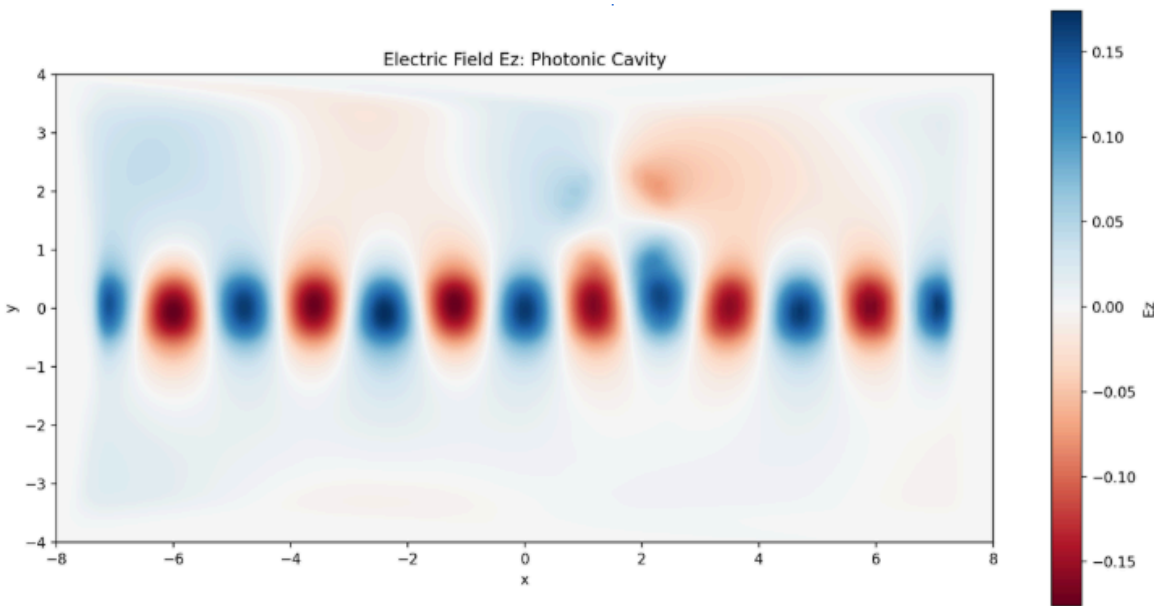
Project 3: Photonic Cavity Simulation and Parameter Optimization

1. Baseline Waveguide



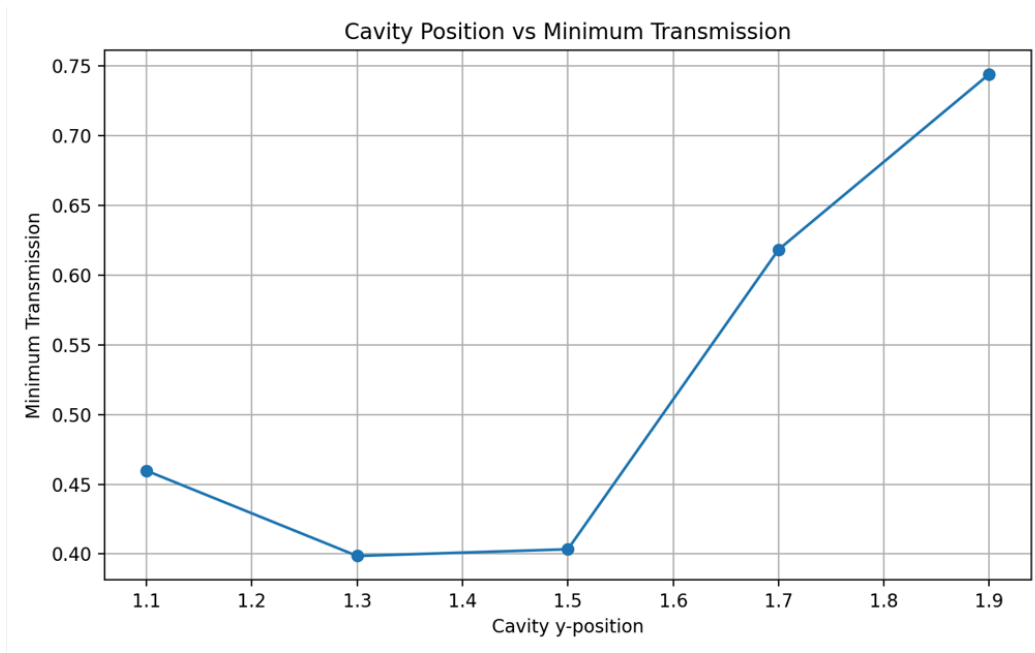
This image shows the electric field E_z propagating through the dielectric waveguide before the photonic cavity is added. It provides the reference field distribution for the rest of the project.

2. Photonic Cavity



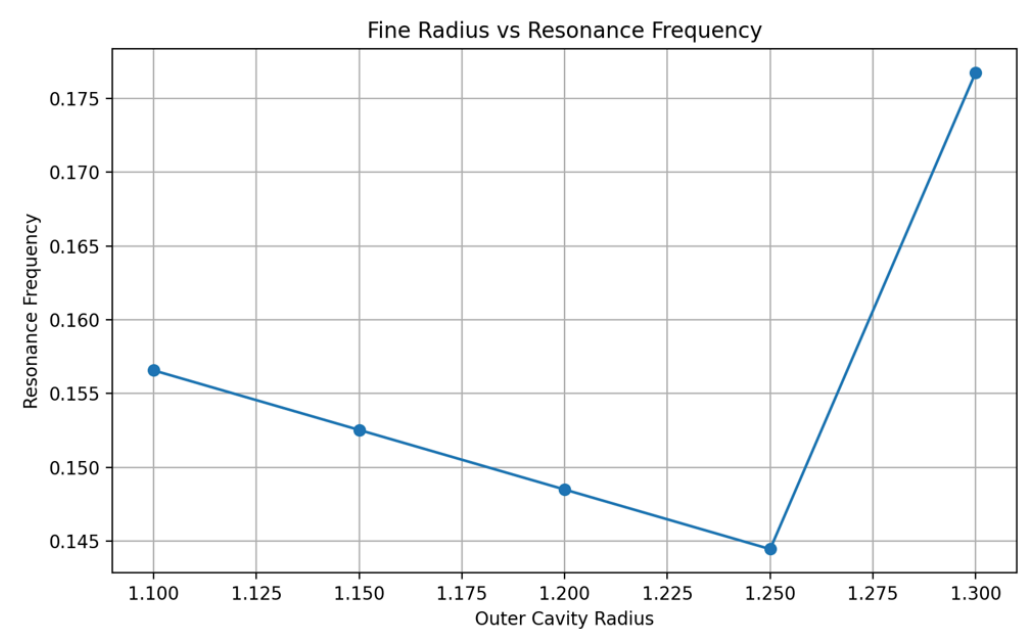
This image shows the electric field E_z after the photonic cavity is introduced. The cavity changes the field distribution near its location.

3. Photonic Cavity Transmission Spectrum



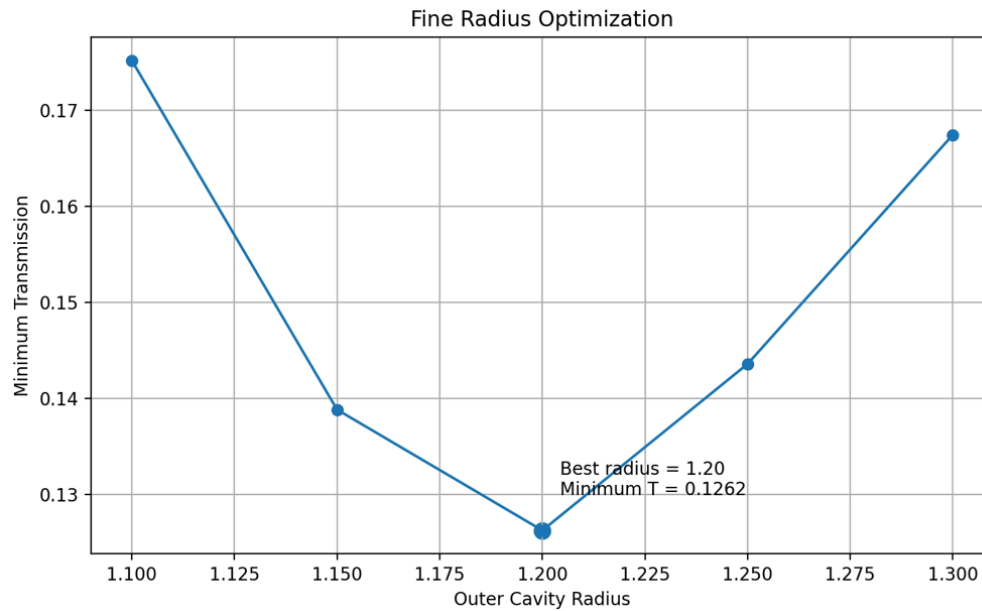
This graph shows transmission as a function of frequency for the photonic cavity. The transmission dips indicate frequencies where the cavity strongly affects the guided wave.

4. Transmission Spectrum vs Cavity Position



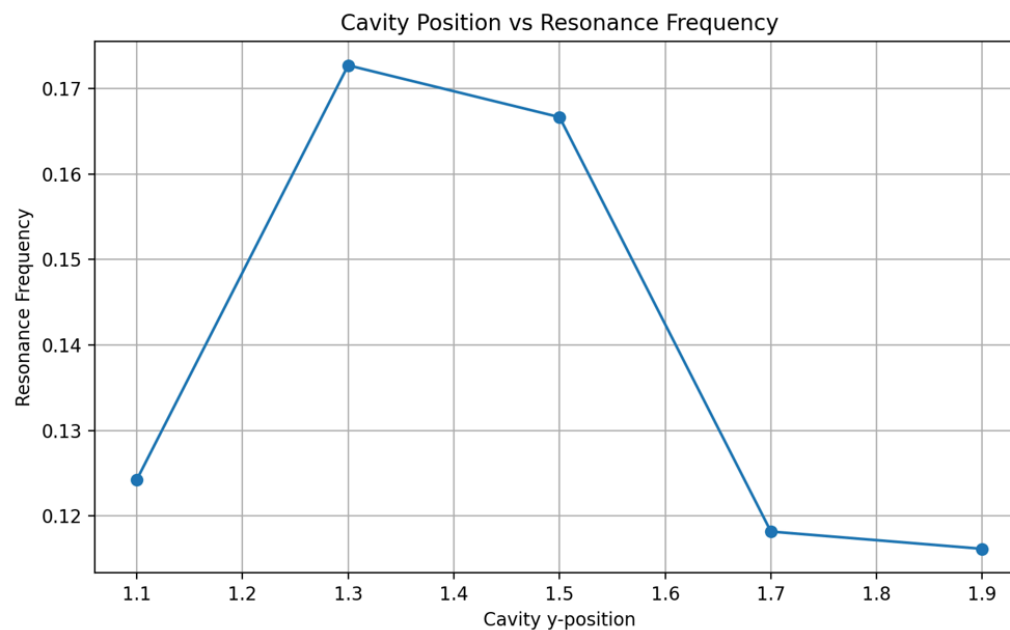
This graph compares the complete transmission spectra for different cavity y-positions. It shows how changing the cavity position affects the frequency response.

5. Cavity Position vs Minimum Transmission



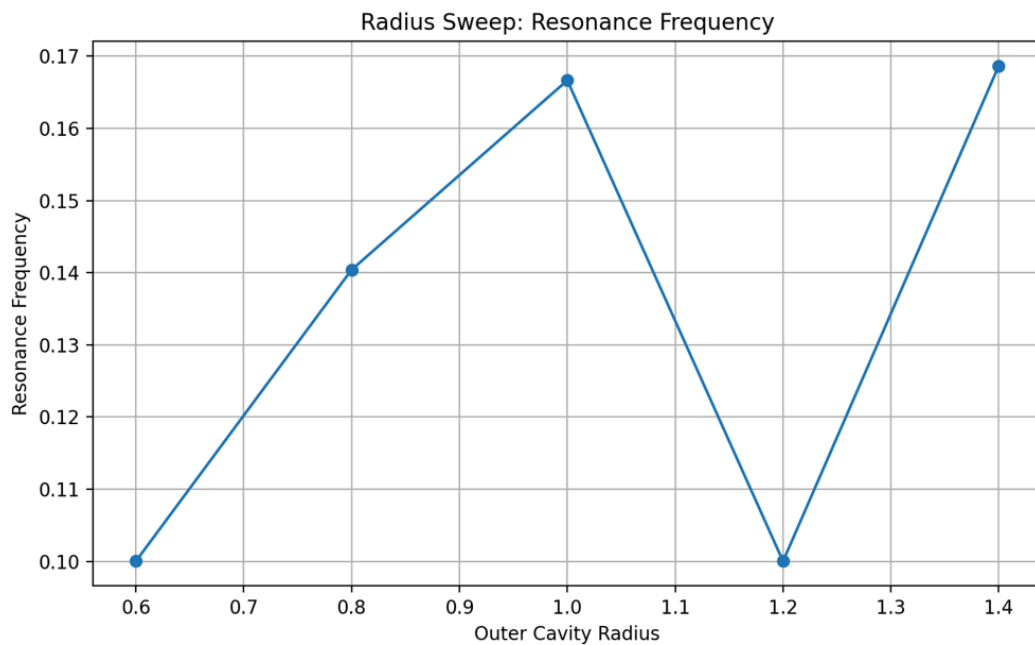
This graph shows the minimum transmission measured for each cavity position. It demonstrates how cavity placement affects coupling strength.

6. Cavity Position vs Resonance Frequency



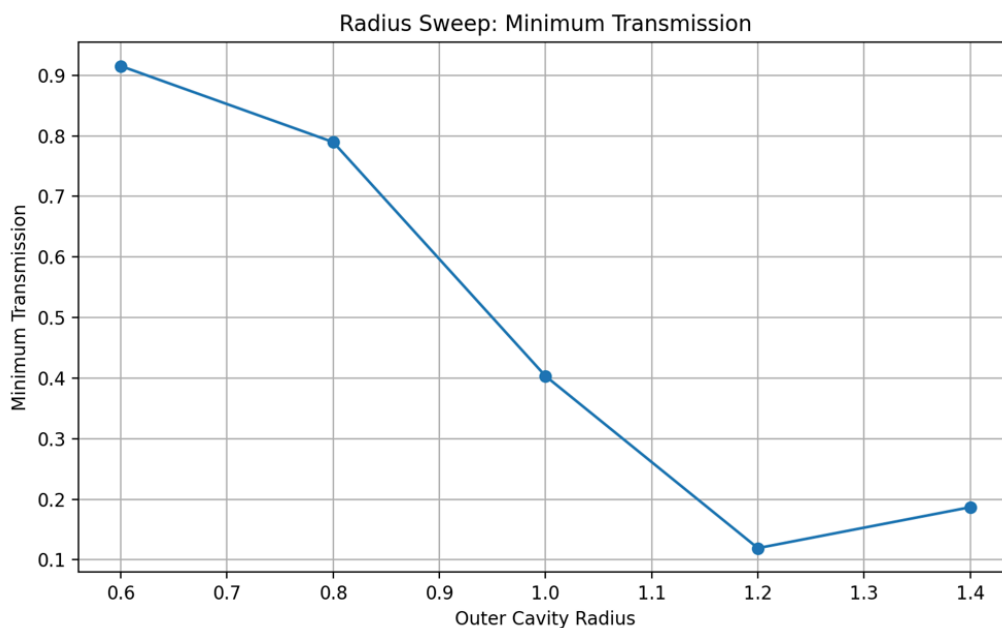
This graph shows how the resonance frequency changes as the cavity position is varied.

7. Transmission Spectrum vs Cavity Radius



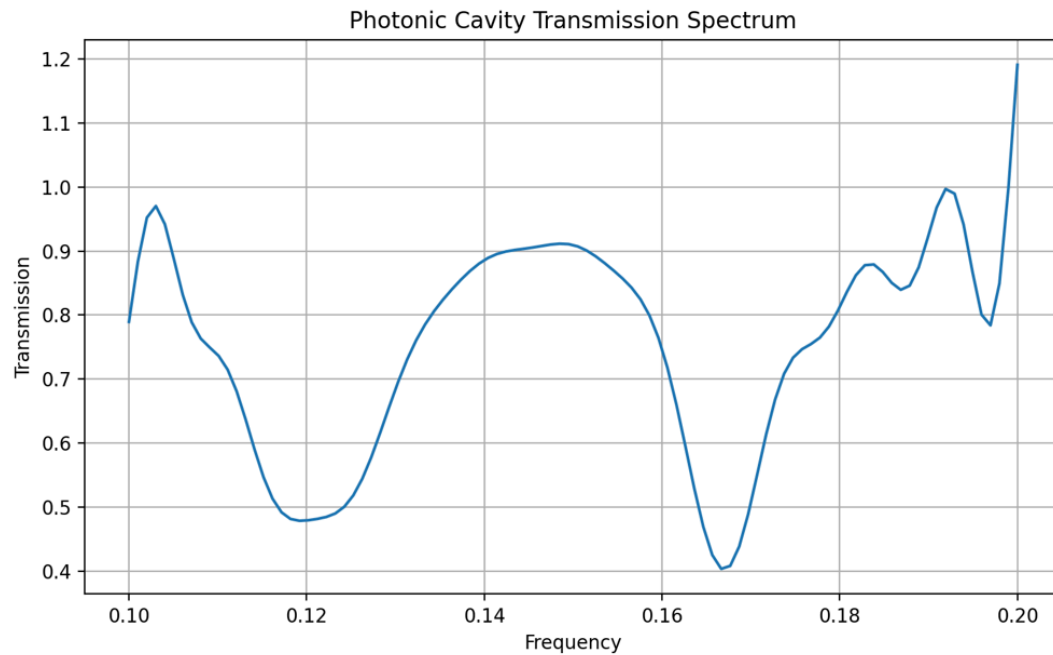
This graph compares the complete transmission spectra for different cavity radii. It shows how changing the cavity size affects the transmission response.

8. Radius Sweep: Minimum Transmission



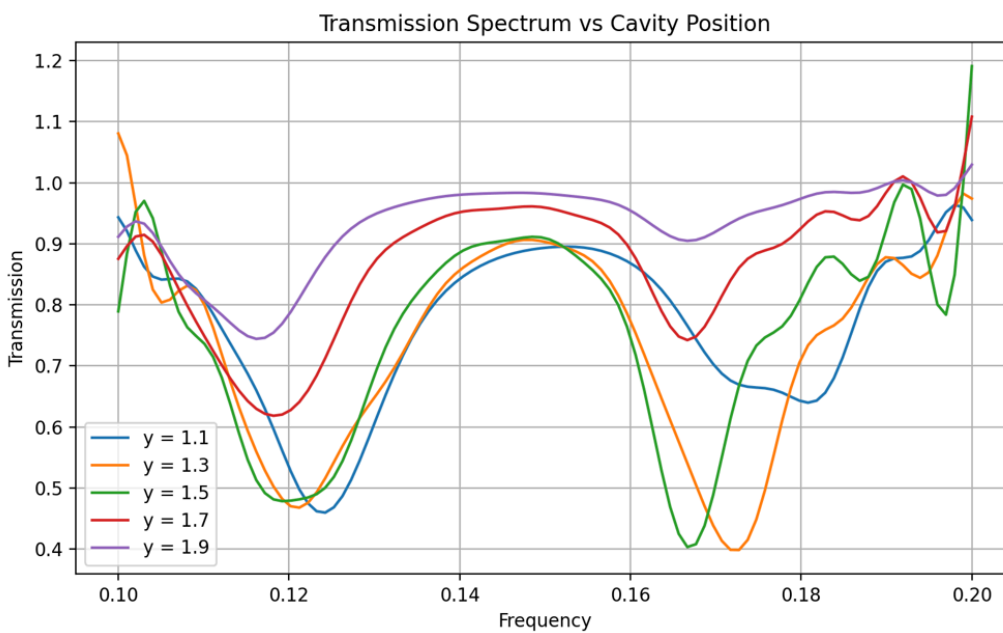
This graph shows the minimum transmission obtained for each tested cavity radius. The results show that cavity radius has a strong effect on transmission suppression.

9. Radius Sweep: Resonance Frequency



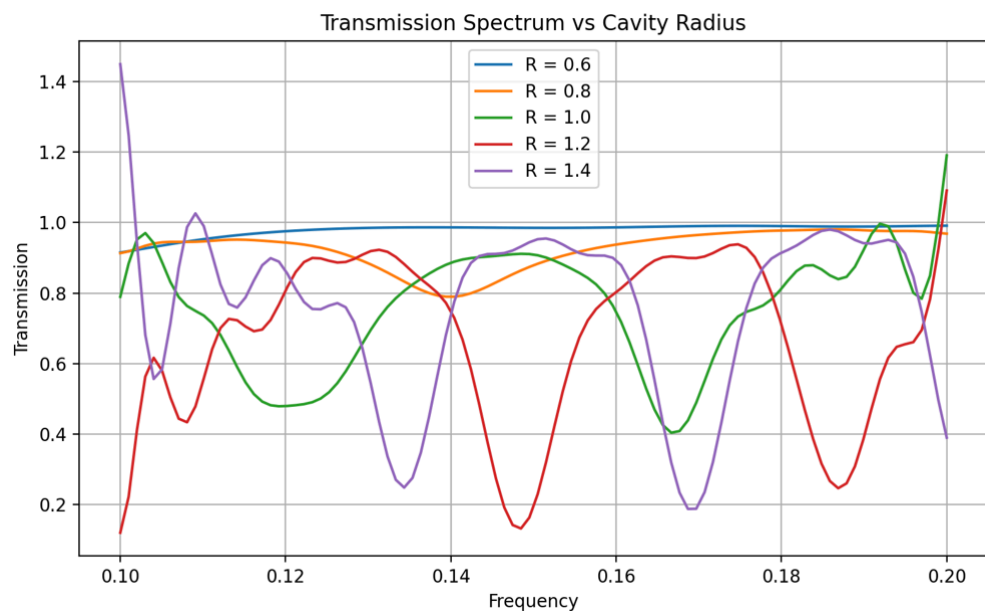
This graph shows how the resonance frequency changes as the outer cavity radius is varied.

10. Fine Radius Transmission Spectra



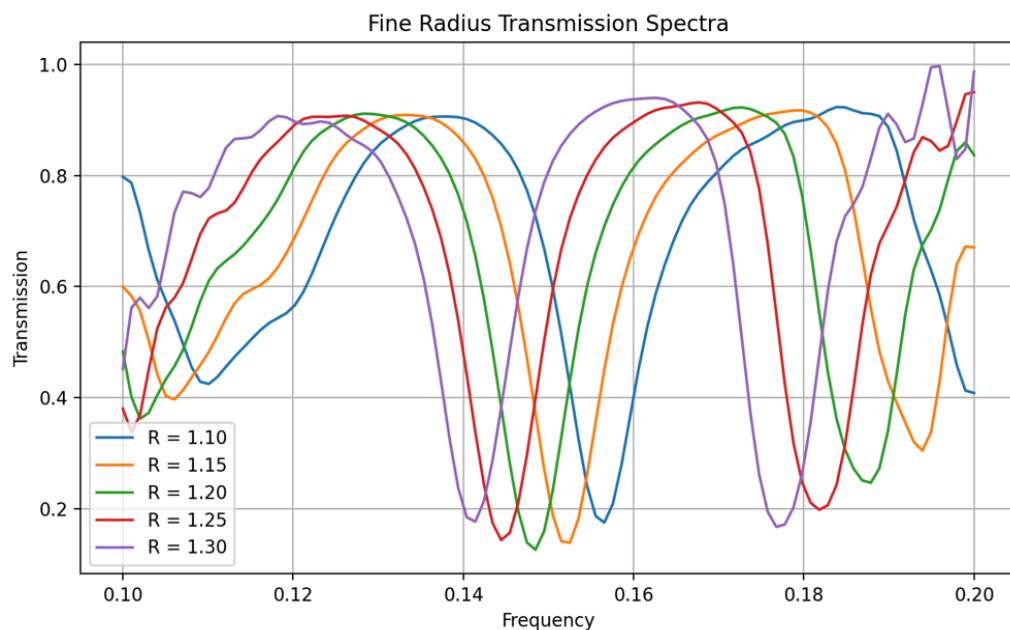
This graph provides a closer comparison of the transmission spectra for the smaller range of cavity radii around the best region identified by the coarse sweep.

11. Fine Radius Optimization



This graph shows the minimum transmission for each radius in the fine sweep and identifies the best tested radius. The minimum transmission occurs at a radius of 1.20.

12. Fine Radius vs Resonance Frequency



This graph shows how the resonance frequency changes across the fine radius sweep.

Final Result

The simulations demonstrate that both cavity position and cavity radius affect the transmission and resonance behavior of the photonic structure. The fine radius sweep identifies a radius of 1.20 as the best tested configuration, with a minimum transmission of approximately 0.1262.